THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME — SET A

Chapters 3 & 4: **The World of Numbers, Exploring Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** terminating	denominator 20 is of the form 2^a5^b
2	**(b)** terminating	denominator 25 is of the form 2^a5^b
3	**(a)** $\frac{3}{11}$	standard method ⇒ $\frac{3}{11}$
4	**(b)** irrational	12 is not a perfect square
5	**(b)** rational	ratio of integers
6	**(d)** $25x^2+20x+4$	$(a+b)^2=a^2+2ab+b^2$
7	**(d)** $25x^2-10x+1$	$(a-b)^2=a^2-2ab+b^2$
8	**(d)** 11025	$105^2=(100+5)^2$
9	**(d)** $(7x+2y)^2$	identify $a^2+2ab+b^2=(a+b)^2$
10	**(b)** $x^2-9x+18$	$(x+a)(x+b)=x^2+(a+b)x+ab$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Average $=\frac{5}{8}$; average of $\frac{1}{2}$ and $\frac{5}{8}=\frac{9}{16}$. So $\frac{9}{16}$ and $\frac{5}{8}$ (any two valid).	1+1
12	$-850+1200-450=-100$: he is in debt of ₹100.	1+1
13	$(x-4)\,(x-3)$	2
14	$(s-8)\,(s-3)$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Multiply numerator and denominator by 5^2 to get 10^3 in the denominator ⇒ exactly 3 decimal places (the last digit is non-zero).	1+2
16	$(n+1)^2-n^2=2n+1=n+(n+1)$ ✓.	1+1+1
17	Removing a debt of ₹5 from 10 rupees of wealth makes you ₹5 richer: $10-(-5)=10+5=15$. In Brahmagupta's language, taking away a debt is like adding a fortune.	1+2
	OR: LHS $=\frac{1}{2}\times\frac{7}{12}=\frac{7}{24}$; RHS $=\frac{1}{8}+\frac{1}{6}=\frac{3}{24}+\frac{4}{24}=\frac{7}{24}$ ✓	1+1+1
18	LHS $=2a^2+2b^2$ ✓. With $a=100, b=5$: $105^2+95^2=2(10000+25)=20050$.	1+1+1
	OR: $104\times96=(100+4)(100-4)=9984$; $98^2=(100-2)^2=10000-400+4=9604$.	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{1}{4}-\frac{5}{12}=\frac{3-5}{12}=-\frac{1}{6}$; $-\frac{1}{6}\div\frac{2}{3}=-\frac{1}{6}\times\frac{3}{2}=-\frac{1}{4}$. (ii) Sum $=\frac{6}{10}+\frac{7}{10}=\frac{13}{10}$; product $=\frac{21}{50}$.	3+2
	OR: (i) No, π is irrational. (ii) $1-\frac{1}{3}+\frac{1}{5}-\frac{1}{7}=\frac{105-35+21-15}{105}=\frac{76}{105}$; times $4=\frac{304}{105}\approx2.895$ — a rough value, improving with more terms. (iii) Lambert, 1761.	1+2+2
20	(i) $(a+b)(a^2+2ab+b^2)=a^3+2a^2b+ab^2+a^2b+2ab^2+b^3=a^3+3a^2b+3ab^2+b^3$. (ii) Replace b by $-b$: $a^3-3a^2b+3ab^2-b^3$. $97^3=(100-3)^3=1000000-90000+2700-27=912673$.	3+2
	OR: (i) $(x-2)(x-3)$; (ii) $(r-7)(r+6)$; (iii) $(2x+1)(3x+2)$; (iv) $(2x-3)(5x+2)$ [since $10x^2-15x+4x-6$].	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapters 3 & 4: **The World of Numbers, Exploring Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** 4	number of places $= \max(a, b)$ in $2^a 5^b$
2	**(d)** -5	Brahmagupta's rules for signs
3	**(a)** division by zero (undefined)	division by 0 is not defined
4	**(d)** $\frac{3}{5}$	distance from 0
5	**(c)** $\frac{1}{7}$	7 is a prime other than 2, 5
6	**(c)** $(x-7)(x+6)$	find a, b with $a+b$ = middle coefficient and ab = constant
7	**(a)** $9s^2 - 25t^2$	$(a+b)(a-b) = a^2 - b^2$
8	**(b)** $p + 2q$	$a = p,\ b = 2q$
9	**(a)** $(x+y)^2 = x^2 + 2xy + y^2$	true for all values of the variables
10	**(c)** Incorrect; the term $2ab$ is missing	$(a+b)^2 = a^2 + 2ab + b^2$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$4 - 15 = -11$ °C.	1+1
12	$4 \times 15 = 60$ and $6 \times 10 = 60$, so $ad = bc \Rightarrow$ equal. (Both equal $\frac{2}{3}$.)	1+1
13	$(x+5)(x+6)$	2
14	$(x-3)(x-2)$	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\frac{x}{3} + \frac{x}{2} = \frac{5x}{6} = \frac{16}{3} \Rightarrow x = \frac{32}{5}$.	2+1
16	Area $= (2x+3)^2 = 4x^2 + 12x + 9$ sq. units; perimeter $= 4(2x+3) = 8x + 12$ units.	2+1
17	Take $OA = 1$; draw a perpendicular at A and mark $AB = 1$. By Baudhāyana–Pythagoras $OB = \sqrt{2}$. With centre O and radius OB, draw an arc to cut the number line at P; P represents $\sqrt{2}$.	1+2
	OR: (i) rational, 9. (ii) irrational. (iii) rational, $\frac{1}{3}$. (iv) rational (repeating block 12345): $\frac{12345}{99999} = \frac{4115}{33333}$. Marking: ½+½+1+1.	½+½+1+1
18	For $a = b = c = 1$: $1 + 1 + 1 = 3$ while RHS $= 6$. So it is NOT an identity.	2+1
	OR: Area $= x^2 + 8x + 15 = (x+3)(x+5)$: sides $x + 5$ and $x + 3$.	2+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) $\frac{126}{10} = \frac{63}{5}$. (ii) $\frac{120}{10000} = \frac{3}{250}$. (iii) $\frac{3052}{1000} = \frac{763}{250}$. (iv) $\frac{23}{99}$. (v) $x = 2.0\overline{5}$: $10x = 20.\overline{5}$, $100x = 205.\overline{5}$, $90x = 185 \Rightarrow x = \frac{37}{18}$.	1+1+1+1+1
	OR: (i) For $a < b$ the average $\frac{a+b}{2}$ is rational and lies between them (3). (ii) $\frac{1}{2}\left(-\frac{3}{12} + \frac{5}{12}\right) = \frac{1}{12}$; it lies to the right of 0, one-twelfth of a unit from the origin (2).	3+2
20	(i) Cubes a^3 and b^3; three cuboids $a \times a \times b$ (vol. a^2b) and three $a \times b \times b$ (vol. ab^2). (ii) $(a+b)^3 = a^3 + 3a^2b + 3ab^2 + b^3$. (iii) $5^3 = 125$; $8 + 3(4)(3) + 3(2)(9) + 27 = 8 + 36 + 54 + 27 = 125$ ✓.	2+1+2
	OR: Multiplying out and cancelling cross terms gives $x^3 + y^3 + z^3 - 3xyz$ (3). Since $1 + 2 - 3 = 0$, the sum $= 3(1)(2)(-3) = -18$ (2). (Check: $1 + 8 - 27 = -18$ ✓.)	3+2

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ANSWER KEY & MARKING SCHEME SET C

Chapters 3 & 4: **The World of Numbers, Exploring Algebraic Identities** | Class 9 Maths | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(b)** non-terminating repeating	denominator 15 has a prime factor other than 2 and 5
2	**(b)** $\frac{2}{3}$	standard method ⇒ $\frac{2}{3}$
3	**(d)** $\frac{1}{6}$	standard method ⇒ $\frac{1}{6}$
4	**(d)** rational	repeating decimal $= \frac{1}{3}$
5	**(d)** irrational	rational + irrational
6	**(d)** $16x^2 + 24x + 9$	$(a+b)^2 = a^2 + 2ab + b^2$
7	**(b)** $9x^2 - 24x + 16$	$(a-b)^2 = a^2 - 2ab + b^2$
8	**(c)** 9801	$99^2 = (100-1)^2$
9	**(b)** $(6x+1)^2$	identify $a^2 + 2ab + b^2 = (a+b)^2$
10	**(c)** $x^2 + 15x + 56$	$(x+a)(x+b) = x^2 + (a+b)x + ab$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	Divide the unit interval $[0, 1]$ into 4 equal parts and move 3 parts to the right of 0.	2
12	$x = 0.\overline{9}$; $10x = 9.\overline{9}$; $10x - x = 9 \Rightarrow x = 1$.	1+1
13	$119^2 = (100+10+9)^2 = 10000 + 100 + 81 + 2000 + 1800 + 180 = 14161$.	2
14	$23 \times 17 = (20+3)(20-3) = 400 - 9 = 391$.	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$15 \div 2 = \frac{15}{2} = 7\frac{1}{2}$: 7 kurtas, with 1 m left over.	1+1+1
16	$x(x-4) = 96 \Rightarrow x^2 - 4x - 96 = 0 \Rightarrow (x-12)(x+8) = 0$. $x = 12$ m (reject −8); breadth 8 m.	1+1+1
17	If $5 - \sqrt{3} = r$ (rational) then $\sqrt{3} = 5 - r$ is rational — contradiction. Hence irrational.	1+1+1
	OR: If $\frac{2}{\sqrt{3}} = r$ (rational, $r \neq 0$) then $\sqrt{3} = \frac{2}{r}$ is rational — contradiction.	1+1+1
18	$r^2 - r - 42 = (r-7)(r+6)$; $x^2 + 7x + 12 = (x+3)(x+4)$.	1½+1½
	OR: $= pqx^2 + (pb+qa)x + ab$. For $(2x+3)(3x+1)$: $6x^2 + (2+9)x + 3 = 6x^2 + 11x + 3$.	1+2

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) -12 and 12. (ii) $-850 + 1200 - 450 = -100$ ⇒ debt of ₹100. (iii) Removing 4 debts of ₹3 each makes one ₹12 richer.	2+2+1
	OR: (i) $\frac{x}{2} = \frac{9}{2} \Rightarrow x = 9$. (ii) LHS $= \frac{1}{2} \times \frac{13}{12} = \frac{13}{24}$; RHS $= \frac{3}{8} + \frac{1}{6} = \frac{9}{24} + \frac{4}{24} = \frac{13}{24}$ ✓	2+3
20	(i) $(1100+4)^2 = 1210000 + 8800 + 16 = 1218816$. (ii) $(200-2)^2 = 40000 - 800 + 4 = 39204$. (iii) $(a+b)^2$ for (i), $(a-b)^2$ for (ii).	2+2+1
	OR: (i) $6x^2 + 11x + 3$. (ii) $96 + 44 + 3 = 143$ m² (check $11 \times 13 = 143$ ✓). (iii) $(x+2)^2 = x^2 + 4x + 4$; $(x+3)(x+1) = x^2 + 4x + 3$ — the square is larger by 1 sq. unit.	2+1+2

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ANSWER KEY & MARKING SCHEME SET D

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SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(c)** -60	Brahmagupta's rules for signs
2	**(c)** $\mathbb{Q}$	Q for quotient
3	**(a)** 1	$x = 0.\overline{9},\ 9x = 9$
4	**(c)** 7	$\lvert -4-3\rvert = 7$
5	**(b)** $\frac{1}{7}$	$\frac{1}{7} = 0.\overline{142857}$
6	**(b)** $(x-9)(x+4)$	find a, b with $a+b$ = middle coefficient and ab = constant
7	**(a)** $4n^2 - 49$	$(a+b)(a-b) = a^2 - b^2$
8	**(d)** 2	$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$
9	**(b)** 40	$2ab = 40$
10	**(d)** 6	$x^2, y^2, z^2, 2xy, 2yz, 2zx$

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	$\left\lvert -\frac{1}{2} - \frac{5}{2}\right\rvert = 3.$	1+1
12	3 decimal places. Multiply numerator and denominator by 5^2 to get denominator 10^3.	1+1
13	$55^2 = 60\times 50 + 25 = 3025$; $65^2 = 70\times 60 + 25 = 4225.$	1+1
14	LHS $= 27$; RHS $= 8+12+6+1 = 27$ ✓.	1+1

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	$\sqrt{1^2+1^2} = \sqrt{2}$; then legs $\sqrt{2}, 1 \Rightarrow \sqrt{3}$; then legs $\sqrt{3}, 1 \Rightarrow \sqrt{4} = 2.$	1+1+1
16	$6\times 2 = 12$; split $7x = 3x + 4x$: $6x^2 + 3x + 4x + 2 = 3x(2x+1) + 2(2x+1) = (2x+1)(3x+2).$	1+1+1
17	$\frac{1}{7} = 0.\overline{142857}$. The block 142857 is a cyclic number: multiplying by 2, 3, … 6 permutes the same digits cyclically.	1+1+1
	OR: $\frac{a+b}{2} - a = \frac{b-a}{2} > 0$ and $b - \frac{a+b}{2} = \frac{b-a}{2} > 0$. So $a < \frac{a+b}{2} < b$; it is rational as sums and quotients of rationals are rational.	1+1+1
18	$(3a)^3 - (2b)^3 = (3a-2b)(9a^2+6ab+4b^2).$	1+2
	OR: $x^2 + \frac{1}{x^2} = \left(x+\frac{1}{x}\right)^2 - 2 = 9 - 2 = 7.$ $\left(x-\frac{1}{x}\right)^2 = x^2 + \frac{1}{x^2} - 2 = 5.$	1+1+1

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	$x = 0.\overline{142857}$, $10^6 x = 142857.\overline{142857}$, $999999x = 142857 \Rightarrow x = \frac{142857}{999999} = \frac{1}{7}$ (3). $2\times 142857 = 285714$, $3\times 142857 = 428571$ — same digits cycled (2).	3+2
	OR: (i) $\lvert -4-3\rvert = 7$; $\left\lvert -\frac{5}{2} - \frac{7}{2}\right\rvert = 6$. (ii) $\frac{7}{9}$, 0, $\frac{5}{3}$. (iii) $\frac{13}{4} = 3.25$ lies between 3 and 4.	2+1+2
20	(i) $x^3 - y^3$ and $x^3 + y^3$. (ii) $(3p+2q)(9p^2 - 6pq + 4q^2)$; $(4-a)(16+4a+a^2)$.	3+2
	OR: $x^2 + \frac{1}{x^2} = 25 - 2 = 23.$ $x^3 + \frac{1}{x^3} = \left(x+\frac{1}{x}\right)^3 - 3\left(x+\frac{1}{x}\right) = 125 - 15 = 110.$	2+3

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